Artix-7 FPGAs Data Sheet: DC and AC Switching - DATASHEET

Manufacturer: AMD Xilinx
Part Number: XC7A200T
Category: FPGA / Low-Power FPGA
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Overview

The Artix-7 FPGA family provides the lowest power and cost solution optimized for high-volume, cost-sensitive applications. Built on 28nm HPL process technology. The XC7A200T features: 215,360 logic cells, 13,140 Kbits block RAM, 740 DSP slices (DSP48E1), 500 I/O pins, up to 16 GTP transceivers at 6.6 Gb/s. Package options: BGA, CSG. Available in commercial and industrial temperature grades.

Power Specifications

Static power (typical): 0.15W (XC7A200T, commercial). Core voltage: 1.0V nominal. I/O voltage banks: 1.2V, 1.5V, 1.8V, 2.5V, 3.3V programmable per bank. XADC: integrated dual 12-bit 1 MSPS ADC for on-chip power and temperature monitoring. Power-on reset (POR) threshold: 0.7V on VCCINT. Recommended decoupling: 100nF per VCCINT pin, 4.7µF bulk per rail.

Key Specifications Summary

logic_cells: 215360. bram: 13140Kb. dsp: 740. io: 500. transceivers: 16. speed: 6.6Gbps. process: 28nm.